Physics-Exact Credit Assignment for Multi-Agent Flutter Suppression

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Differentiating a structural plant's aeroelastic energy along its trajectories isolates a control-power term that is exactly additive across an arbitrary number of control inputs. We use this to give multi-agent reinforcement learning, applied to distributed flutter suppression on independently actuated surfaces of a Goland wing, a closed-form per-agent credit signal in place of the estimated one the method normally requires. Combined with log-energy shaping, this credit beats the closed form alone, a naive reward, and an adapted value-decomposition baseline, but not shaping alone once corrected for the full comparison family. Its advantage over the estimated alternatives is reliability, not raw mean performance, at higher variance. None of this was visible at a conventional exploration setting: the useful actuation amplitude is roughly forty times smaller than a conventional policy's exploration noise, and matching exploration to it changed performance by a factor of two. The credit signal tolerates plausible identification error, and a classical decentralised controller's fragility at the edge of the flight envelope is not a tuning artefact. The learned policies still trail that controller on average damping while holding one flight condition it cannot, pointing toward a certified, monitored-residual architecture not yet built.

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Nomenclature

$q, \dot{q}, \ddot{q}$	modal coordinates, rates, accelerations
M, C, K	modal mass, damping, stiffness matrices
L, x_w	wake-coupling matrix and wake (Wagner lag) states
B, δ, b_k	control-influence matrix, deflection vector, its kth column
E	instantaneous structural energy, $\frac{1}{2}\dot{q}^\top M\dot{q} + \frac{1}{2}q^\top Kq$
P_k	control power of agent k, $(\dot{q}^\top b_k)\delta_k$
λ	energy decay rate, $\frac{1}{2}\,\mathrm{d}(\log E)/\mathrm{d}t$
D_k	difference reward for agent k
U, U_f	airspeed, flutter speed
N	number of control surfaces / agents
R	number of retained modal phasors (R=2 throughout)
o_k, a_k	agent k's local observation and normalised action
$\hat{\phi}_k$	agent k's encoded phasor estimate, $\in \mathbb{R}^{2R}$
e_k	consensus message received by agent k
σ	initial policy exploration standard deviation (normalised action units)
d, p	Cohen's d effect size, Welch's t-test p-value

I. Introduction

Flutter is a self-excited aeroelastic instability in which structural bending and torsion couple through unsteady aerodynamic loads and draw energy from the freestream; past a critical speed the response grows without bound within a few cycles, with no warning a pilot can act on. Because a wing must be certified against flutter with a substantial speed margin, the mass, stiffness and manufacturing cost of the structure are set in large part by an instability it is never meant to encounter in service [1, 2]. Active suppression buys back some of that margin with control effort rather than structural weight, and has attracted sustained engineering attention since the 1970s [3–5].

Two trends motivate revisiting the problem now. Wings are becoming more flexible and carry more independently actuated trailing-edge surfaces, each with its own local sensing on a bus rather than a dedicated wire to a central controller: the hardware is already distributed even where the control law is not. Reinforcement learning has also been applied to flutter suppression with encouraging results [6–8], but every instance we are aware of uses a single agent commanding a single effector from a global observation. Taking the hardware trend seriously means asking what happens when the effectors are commanded by separate agents that cannot see the whole wing, which is the question this paper answers on a specific, validated testbed (Figure 1).

Posing the problem with one agent per surface makes credit assignment the central difficulty, and for a physical reason: every surface acts on the *same* coupled bending–torsion mode, so a reward built from the whole wing's response says little about which surface helped. Standard multi-agent reinforcement learning remedies estimate that decomposition statistically from returns a policy happens to collect [9–11]. Our first observation is that this class of plant does not require estimation at all: for a structural system whose energy budget is exactly additive across control inputs, the credit decomposition is available in closed form from the equations of motion themselves.

Our second observation is the one we would most want a practitioner to take away. Before finding it, we compared credit signals, communication schemes, action parameterisations, recurrence and auxiliary losses, and every variant plateaued in the same narrow performance band regardless of mechanism — evidence that something *upstream* of all of them was binding. The policy's exploration standard deviation was roughly forty times the deflection amplitude that actually damps this wing, so no credit signal, however well constructed, could assign credit to a signal buried that far under noise. Matching exploration to the useful actuation amplitude changed performance by a factor of two, more than any algorithmic choice we tried, and only then did the credit-assignment effect we set out to study become measurable at all.

This echoes a known failure mode: the reinforcement learning literature has repeatedly found implementation and hyperparameter choices dominating algorithmic ones in general-purpose benchmarks [12–14]. Our contribution is a control-specific instance of that caution, with a diagnostic usable *before* training: the useful actuation amplitude is computable from a classical design, and the exploration scale can be checked against it in advance rather than discovered by trial and error.

Contributions

1) A closed form for the multi-agent difference reward in an energy-conserving structural plant with additive control authority (Proposition 1), together with the correction that makes it usable: the credit must be the logarithmic decay rate, not the raw energy rate (Section IV.B).
2) A five-variant credit-assignment comparison, corrected for the number of comparisons made, against a naive undecomposed reward and an adapted value-decomposition baseline in the style of VDN as well as against each half of the closed-form-plus-shaping credit alone: the closed-form credit's advantage over an estimated one is in consistency across seeds and conditions, not a uniformly larger effect size (Section V.G).
3) A quantified account of what limits learning on this problem, with a diagnostic a practitioner can apply in advance, and a capability probe that rules out the competing explanation (Section V.I), together with a direct

check of how much control-influence identification error the credit signal's sign tolerates (Section V.J).

4) A validated testbed and a six-rung classical baseline ladder that separates the control method from the information structure available to it, cross-validated against an unsteady vortex-lattice solver, including a check of whether the classical baseline's fragility at the envelope edge is a tuning artefact rather than an assumption (Sections III.C, V and V.F).
5) A negative result reported as such: the communication and action parameterisation mechanisms we derived from the physics do not help, at a level of statistical correction that makes the negative result harder to dismiss rather than easier.

II. Related work

Active flutter suppression The classical approach synthesises a single centralised regulator from a state-space aeroelastic model, driving all control surfaces from one observer and one gain [3]. Robust multivariable designs following that template have been demonstrated on wind-tunnel models [4] and extended to nonlinear aeroelastic structures with limit-cycle behaviour [5]; Livne [1] surveys the gap between laboratory demonstrations and certified hardware, and Mukhopadhyay [2] traces the field's history from Theodorsen to modern aeroservoelastic synthesis. None of this literature questions the centralised structure itself: the regulator sees the whole state and commands every surface jointly, an assumption a physically distributed set of independently wired trailing-edge surfaces need not satisfy. Karpel's original formulation [3] treats gust alleviation and flutter suppression as the same state-feedback problem, a framing we inherit for the objective in Section III even though our control structure departs from it. The unsteady aerodynamic models underlying both the classical literature and our own plant descend from Theodorsen [15] and the indicial (Wagner-function) formulation of Jones [16]; finite-state induced-flow models [17] extend the same lag-state idea to a richer wake representation than the two-state approximation we use, and [18–20] are the standard texts our modelling conventions draw from.

Adaptive flutter suppression A mature adaptive-control literature sits between the fixed-gain designs above and the learned policies below, aimed at the gap our own motivation cites: an aerodynamic model that is never exactly the design model in flight. Downs and Prazenica [21] apply direct adaptive control to a flexible wing with nonlinear stiffness and actuator freeplay, stabilising post-flutter conditions a fixed-gain LQR cannot; Mozaffari-Jovin et al. [22] design an $\mathcal{L}_1$ adaptive law for the same flap-actuated setting we study. Both retain a single, centralised regulator and adapt its parameters online rather than decomposing the effectors into separate agents, answering a different question from ours — robustness to model error at one control point, not what changes when the control points themselves are separate decision-makers.

Learning for flutter control A recent line of work applies deep reinforcement learning directly to flutter suppression: stall flutter via active camber morphing [7], trailing-edge circulation control validated in a wind tunnel [6], and tuning a conventional aeroservoelastic controller's parameters [8]. Despite different effectors and plants, all three share the same structural assumption as the classical literature they replace a hand-designed gain in: one learning agent, one effector, one global observation. None asks what changes when the effectors are commanded by separate agents that cannot see the whole wing, which is the gap this paper occupies. A recent follow-up to the camber-morphing line distills a trained single-agent controller into a transparent, rule-based policy via Shapley additive explanations [23] — a credit-assignment concept applied *post hoc* to interpret an already-fitted network, unlike our closed-form decomposition of the reward several agents train against, derived from the plant's own energy budget before training begins. Framing a wing's surfaces as a multi-agent system is not without precedent: Shalymov et al. [24] synthesise a speed-gradient adaptive control law across many small actuated "feathers," which is multi-agent *control* in the classical sense rather than separate *learning* agents estimating from partial observability, the distinction that gives rise to the credit-assignment question this paper poses.

Decentralised control That decentralisation costs closed-loop performance relative to a centralised design with the same model is a classical result, not a symptom of using learning [25, 26]. Bakule's survey [26] catalogues the design patterns this paper's baseline ladder instantiates and reports the same qualitative cost across domains well outside aeroelasticity. Rotkowitz and Lall [27] characterise when decentralised synthesis remains *tractable*: convex synthesis survives decentralisation exactly when the information structure is *quadratically invariant*, a condition our line-graph sensing pattern does not satisfy, which is why the fully decentralised LQG baseline (Supplemental Material, Section S4) is synthesised independently per surface. This literature is why we treat the information structure as an experimental variable: the honest classical baseline for a policy that sees only local accelerometers is a fully decentralised LQG built from the same sensing, not a centralised design the learned policy could never match.

Multi-agent credit assignment The standard tools for the credit-assignment problem we pose are difference rewards [28] and learned value factorisation [9, 10], with COMA [11] using a counterfactual advantage baseline; MADDPG [29] and MAPPO [30] are common actor-critic baselines built on these ideas, surveyed more broadly in [31, 32], with estimators still being refined [33]. All of these estimate the credit decomposition statistically from returns a policy happens to collect, because the environment is treated as a black box. Our contribution is narrower and complementary: for the energy-conserving structural plant studied here, the decomposition these methods estimate is available in closed form directly from the equations of motion (Proposition 1), with no learning, counterfactual rollout, or critic involved. We see this as evidence that physical structure, where it exists, is worth exploiting before reaching for a general estimator built to work without it. The formal setting for the distributed problem is the decentralised POMDP

[34, 35], whose complexity results motivate treating partial observability as intrinsic to the sensing architecture rather than a simplification to be relaxed.

Learned communication CommNet [36] and TarMAC [37] learn message content end to end, with TarMAC also learning *who* to attend to. We instead fix message content to a physics-derived modal phasor and message routing to the spanwise line graph, so the channel stays interpretable and bandwidth-bounded independently of agent count — a property that matters on a physical avionics bus. Neither the fixed content nor the fixed routing helped relative to a plain policy with no channel at all (Section V.G); we report this negative result rather than omit the mechanism.

Multi-agent benchmarks and physical grounding The environment is implemented against the PettingZoo parallel-environment API [38], the standard interface much of the literature above is evaluated through, so the mechanisms surveyed above drop into it largely unmodified, making the ablations of Section V.G a fair test of those mechanisms rather than of our own re-implementation. Unlike the particle- and grid-world benchmarks that dominate the credit-assignment literature, this environment's dynamics are a validated model of a physical instability rather than authored for the coordination problem they pose, which we offer as one instance of a currently under-represented axis: physically-grounded multi-agent control tasks.

Reinforcement learning practice Our central caution has direct precedent in general-purpose reinforcement learning, though not, to our knowledge, applied to a physical control problem where the binding quantity is predictable in advance. Henderson et al. [12] documented seed-to-seed variance swamping algorithmic differences; Engstrom et al. [13] showed implementation details, not the nominal algorithm, accounted for much of PPO's reported advantage over TRPO; Andrychowicz et al. [14] found hyperparameters dominating architectural choices at scale; and Agarwal et al. [39] argued for reporting effect sizes and uncertainty rather than point estimates, a practice we follow throughout Section V.G. Yu et al. [30] make the multi-agent analogue: much of MAPPO's reported advantage traces to implementation details rather than algorithmic novelty, part of why Section IV.D documents our own architecture choices at the level of individual design decisions. What we add is a case in which the dominant hyperparameter — the exploration scale — is *computable in advance* from a classical design of the same plant, unlike hyperparameters in the general-purpose benchmarks this literature studies. Residual formulations, in which a learned policy corrects a fixed classical controller rather than replacing it [40, 41], appear here as one of the variants we evaluate.

Positioning Table 1 summarises the four axes above together, since no single prior study we are aware of combines more than one of them: multi-agent treatment of the individual control surfaces of one wing, a credit signal derived in closed form from the plant's own physics rather than estimated, validation of the plant itself against a higher-fidelity solver rather than only against a published linear benchmark, and an explicit, reproducible account of what a conventional

Table 1 This paper against the literature it draws on, along the axes Section II discusses. A ✓ marks that the property holds for the representative citation(s) in that row; – marks that it does not; `n/a` marks that the axis does not apply to that line of work at all (a single-agent or single-regulator method has no multi-agent credit decomposition to be closed-form *or* estimated, so neither ✓ nor – would be meaningful there).

	Multi-agent surfaces	Closed-form credit	Validated against higher-fidelity solver
Classical AFS [3, 4]	–	n/a	varies
Adaptive AFS [21, 22]	–	n/a	–
Learned AFS [6–8]	–	–	–
General MARL credit [9–11]	✓	–	n/a
Residual control [40, 41]	–	–	n/a
This paper	✓	✓	✓

exploration setting hides. Each axis is individually present somewhere in the literature surveyed above; their combination, and the negative and cautionary results that combination surfaces, is what this paper contributes.

III. Problem formulation and plant model

A. Decentralised partially observable decision process

We model the task as a Dec-POMDP [34, 35] $\langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_k\}, T, \{\Omega_k\}, O, R, \gamma \rangle$. The agent set $\mathcal{I} = \{1, \ldots, N\}$ has one member per control surface. The state $s = (q, \dot{q}, x_{\mathrm{w}}, \delta)$ collects the modal coordinates, their rates, the wake states and the current deflections. Agent k chooses a normalised command $a_k \in [-1, 1]$, which its actuator converts to a physical deflection subject to a rate limit and travel stops.

Each agent observes only $o_k \in \Omega_k$: an accelerometer pair at its own station, local plunge rate, twist and twist rate, its own saturation fraction, and air data. Crucially o_k does not contain the modal state. The unstable mode is a global quantity that no single station can measure, so the partial observability is intrinsic to the physical arrangement rather than an artefact of our formulation. Communication, where used, is nearest-neighbour along the span.

B. Objective

We use the standard discounted formulation [42]. The quantity to be minimised is the exponential growth rate of the structural energy E, which is also the quantity we report:

$$\lambda = \tfrac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \log E, \tag{1}$$

negative when the wing is being damped. Stating the objective this way, rather than as an accumulated quadratic cost, matters later: it is what allows the per-agent credit and the evaluation metric to be literally the same quantity (Section IV.B).

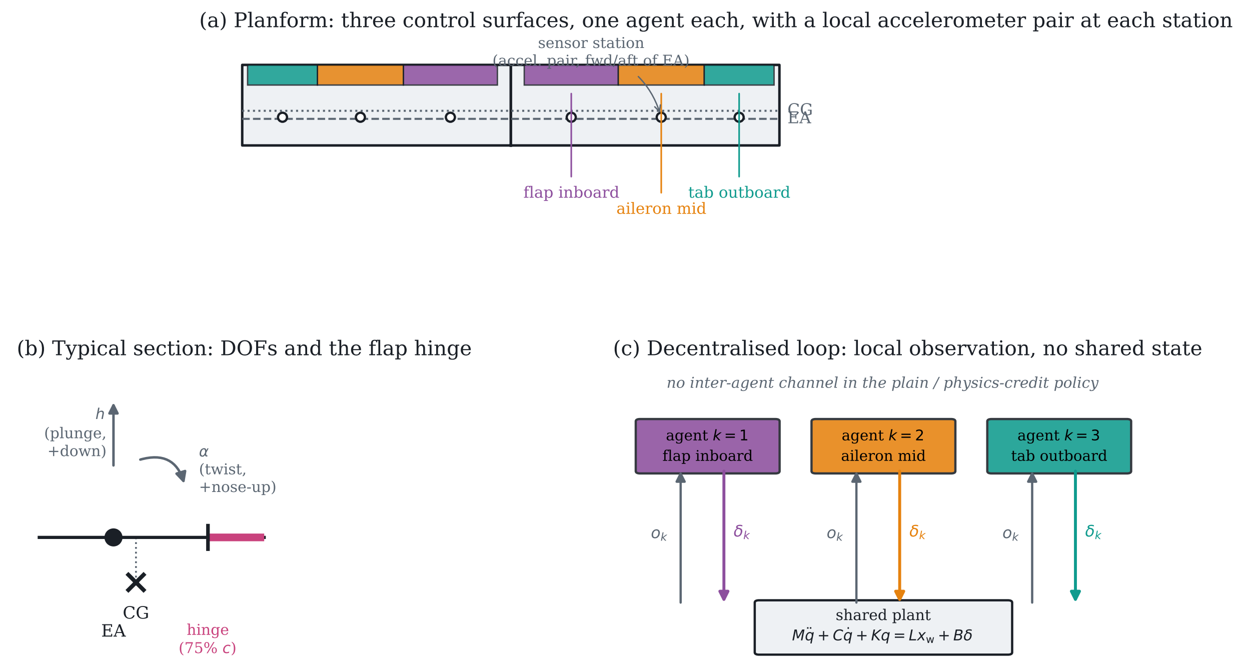

Fig. 1 The testbed. (a) Planform, agent colours matched throughout the paper. (b) The typical-section degrees of freedom and the flap hinge that the thin-airfoil derivative in Section III.F refers to. (c) The information structure: each agent closes its own loop through the shared plant, with no inter-agent channel in the plain and physics-credit policies.

C. Model

We use a modally reduced uniform cantilever wing, shown schematically in Figure 1. Three trailing-edge surfaces occupy spanwise bands at $[0.05, 0.40]$, $[0.40, 0.72]$ and $[0.72, 0.98]$ of the semispan (panel a), hinged at 75 % chord (panel b), with $\pm 15^\circ$ travel, $200\,^\circ\,\mathrm{s}^{-1}$ rate limit and a 20 ms first-order servo lag. Each surface is commanded by its own agent, which observes only a local accelerometer pair, local plunge and twist rate, its own saturation fraction, and air data; no agent observes the modal state, and the three agents share no channel unless a communication mechanism is explicitly under test (panel c). Control runs at 200 Hz.

D. What flutter and its suppression look like

Every quantity in the rest of the paper is a scalar derived from a wing that is physically bending and twisting, worth seeing once before reducing it to decay rates. Figure 2 gives the same initial 5 cm tip pluck under two treatments at $U/U_\mathrm{f} = 1.10$, six shared instants apart, on one shared deformation scale. The top row is what "flutter" names: bending and torsion lock into a fixed phase relationship and the response grows visibly within under a second, until the linear model is no longer valid. The bottom row is the identical pluck under centralised LQR: by the second frame the deflection is already smaller than at $t = 0$, and stays there. Both are literal rollouts of the plant in Section III.C at the flight condition used throughout Section V.E. A supplementary animation of all four classical cases is provided

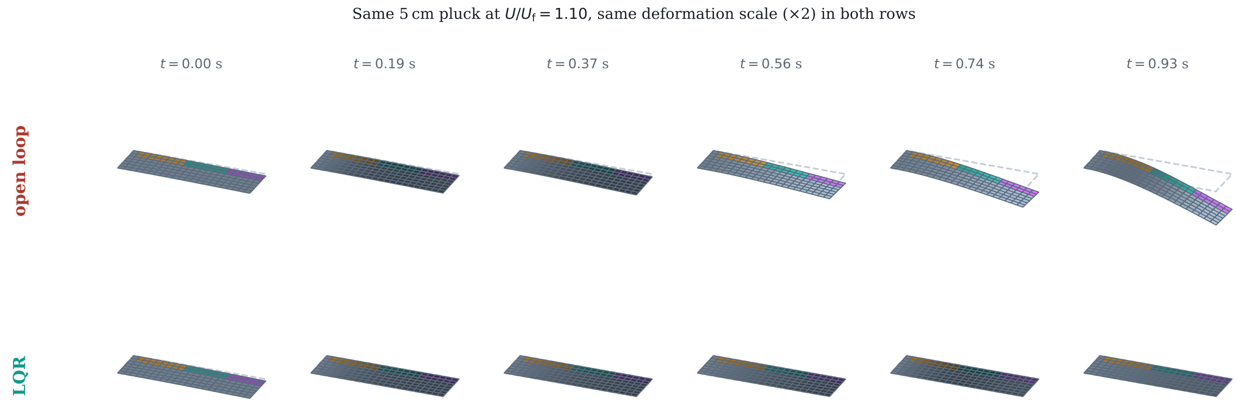

Fig. 2 Six shared instants of the same 5 cm **initial tip pluck at** $U/U_{\mathrm{f}} = 1.10$**, one shared deformation scale (dashed outline: the undeformed reference planform). Top: open loop – bending and torsion coalesce and the response diverges within** 0.93 s**, at which point it leaves the linear regime and the rollout is stopped. Bottom: the identical pluck under centralised LQR, suppressed by the second frame and held near zero afterward. Agent colours match Figure 1.**

alongside the code (Data and Code Availability).

The structure is a Galerkin projection onto exact clamped-free bending and fixed-free torsion modes, with all generalised mass and stiffness integrals evaluated by Gauss–Legendre quadrature. Aerodynamics are strip theory carrying the full Theodorsen non-circulatory terms [15], with the circulatory lag realised in the time domain by two Wagner lag states per strip [16]. Control surfaces contribute quasi-steady thin-airfoil flap derivatives from the Glauert series, integrated over each surface's spanwise band. The equation of motion is

$$M\ddot{q} + C\dot{q} + Kq \;=\; L\,x_{\mathrm{w}} + B\delta + b_{\mathrm{g}}w_{\mathrm{g}}, \tag{2}$$

with plunge positive down and twist positive nose-up, following the sign convention shown in Figure 1b. The state has 56 dimensions: four modal coordinates and their rates, plus two wake states on each of 24 strips. Full geometric, structural and actuator parameters are tabulated in A for reproducibility.

E. Validation against the published benchmark

Table 2 compares the model against the Goland cantilever wing [43]. Agreement is better than 0.5 % on the modal frequencies and the flutter speed, and this is enforced by an automated test against published values rather than against numbers the implementation once produced.

Table 2 Validation against the published Goland wing.

Quantity	This model	Published
First bending frequency	7.664 Hz	7.66 Hz
First torsion frequency	15.232 Hz	15.24 Hz
Flutter speed	$137.35\,\mathrm{m\,s^{-1}}$	$\approx 137\,\mathrm{m\,s^{-1}}$
Flutter frequency	$69.3\,\mathrm{rad\,s^{-1}}$	$\approx 70.7\,\mathrm{rad\,s^{-1}}$
Static divergence speed	$356.8\,\mathrm{m\,s^{-1}}$	—

F. Cross-validation against an unsteady vortex-lattice solver

The model buys speed with two approximations a reader is entitled to distrust: two-dimensional strip aerodynamics, and quasi-steady flap derivatives that neglect the flap's own shed wake. We quantified both against SHARPy [44], which couples an unsteady vortex-lattice method [45] to a geometrically exact beam. The comparison is clean because SHARPy's own Goland template carries structural data identical to ours, so any disagreement is about the aerodynamics.

Flutter point SHARPy places flutter at $154.46\,\mathrm{m\,s^{-1}}$ and $69.3\,\mathrm{rad\,s^{-1}}$ at sea level, against $137.35\,\mathrm{m\,s^{-1}}$ and $69.34\,\mathrm{rad\,s^{-1}}$ for our model. The *frequency* agrees to 0.06 %, so both identify the same bending–torsion coalescence; the critical speed differs by −11.1 %, the direction two-dimensional strip aerodynamics predicts, since it has no tip relief, over-predicts loading near the tip, and therefore under-predicts the flutter speed. Our plant is conservative, the safe direction for a control study.

Control authority This discrepancy runs the other way and is larger. A part-span flap loses effectiveness to spanwise carryover and to flow around its ends, neither of which a sectional derivative represents. Integrating our thin-airfoil $C_{l\delta} = 3.826$ per radian over the flapped span (25 % of span, read from the generated aerodynamic grid) gives a whole-wing $\mathrm{d}C_L/\mathrm{d}\delta$ of 0.957 per radian, against 0.495 from UVLM. Our plant over-estimates control authority by a factor of 1.9, grid-converged to within 3 % between 8×16 and 16×32 panels.

Consequence Every controller in this study, classical and learned, was evaluated on the same plant and received the same inflated authority, so relative comparisons, ablations and significance tests are unaffected. Absolute decay rates would not transfer unchanged to a UVLM plant, and we do not claim they would. The two discrepancies act in opposite directions — a harder flutter problem, an easier control problem — so they cannot be combined into a single safety factor.

IV. Method

A. An exactly additive energy budget

Differentiating the aeroelastic energy $E = \frac{1}{2}\dot{q}^\top M\dot{q} + \frac{1}{2}q^\top Kq$ along trajectories of (2) gives

$$\dot{E} = \underbrace{-\dot{q}^\top C\dot{q}}_{\text{dissipation}} + \underbrace{\dot{q}^\top Lx_{\mathrm{w}}}_{\text{wake exchange}} + \sum_k \underbrace{(\dot{q}^\top b_k)\,\delta_k}_{P_k}, \tag{3}$$

with b_k the kth column of B. The final term is exactly additive across agents; $P_k < 0$ means surface k removes energy from the structure.

Proposition 1 (Closed-form difference reward)**.** *For a global objective whose only agent-dependent term is the control power in* (3)*, the difference reward $D_k = G(z) - G(z_{-k})$ admits the closed form $D_k = -P_k$. No counterfactual rollout, learned mixing network or centralised critic is required to evaluate it.*

Proof. Removing agent k sets $\delta_k = 0$. Every other term of (3) is independent of δ_k, and the control-power term is a sum over agents, so the difference collapses to $(\dot{q}^\top b_k)\delta_k$. □

Projecting onto the in-vacuo modes keeps the separation in agent *and* mode: agent k's contribution to mode r is $\dot{\eta}_r(v_r^\top b_k)\delta_k$. The factor $v_r^\top b_k$ is the modal residue, which decides whether a surface damps or excites that mode and whose sign inverts under control reversal.

B. The credit must be the logarithmic decay rate

Used directly, $-P_k$ fails instructively. It is maximised by having energy available to extract, so an agent is rewarded for *sustaining* the oscillation. Policies trained on it held the wing at a bounded but undamped amplitude, scoring $+0.265\ \mathrm{s}^{-1}$ against -0.06 for a hand-tuned collocated damper. This is reward hacking, not myopia.

Dividing (3) by E repairs it:

$$\frac{\mathrm{d}}{\mathrm{d}t}\log E = \frac{-\dot{q}^\top C\dot{q} + \dot{q}^\top Lx_{\mathrm{w}} + \sum_k P_k}{E}, \qquad r_k = -\frac{P_k}{E + \varepsilon}. \tag{4}$$

Because E is a shared scalar the decomposition survives division exactly, so Proposition 1 carries over to the log-decay rate. The signal is now scale invariant, and it coincides with the evaluated objective (1). Making the change inverted the sign of the outcome, from $+0.265$ to $-0.291\ \mathrm{s}^{-1}$, with training divergence falling from 0.8 % to 0.0 %.

Remark 1. *The exact signal remains myopic: cross-mode coupling through C and through the wake means an agent may have to inject energy into one mode so another can extract more from the coupled pair. We therefore add potential-based shaping [46] on* $\log E$*. We use a shaping discount of one rather than matching the RL discount, which forfeits the strict*

Algorithm 1 Training with the blended (exact-plus-shaping) credit

Assemble M, C, K, L, B at the sampled airspeed; small-gain init of the action head
for each update **do**
  Ease the task distribution by the curriculum fraction
  for $t = 1, \ldots, T$ **do**
    Each agent k acts on its own o_k^t; step the coupled plant and actuators
    $P_k^t \leftarrow (\dot{q}^\top b_k)\, \delta_k^t$ ▷ closed form, Prop. 1
    $r_k^t \leftarrow -P_k^t/(E^t + \varepsilon) + w\,[\Phi^{t+1} - \Phi^t]/\Delta t$ ▷ $\Phi = -\log \frac{E+\varepsilon}{E_{\text{ref}}}$
  end for
  Compute per-agent GAE advantages
  for each epoch, each minibatch of *sequences* **do**
    Unroll L_{bptt} steps from a detached hidden state; zero it where an episode ended
    PPO clipped objective + value loss − entropy bonus
  end for
end for

policy-invariance guarantee. The reason is concrete: with a discount below one the leakage term $(\gamma - 1)\Phi/\Delta t$ becomes a constant negative reward once the wing is damped — measured at -6.91 per step, totalling -977 over an episode for a controller that damps perfectly, against roughly zero for one that lets the wing ring. An invariance theorem is of no use if the shaped reward ranks a ringing wing above a damped one.

C. Training

We use shared-parameter multi-agent PPO [30, 47] with generalised advantage estimation [48] and per-agent advantages, since each agent receives a different reward. An on-policy method is preferred here over off-policy alternatives [49, 50] because the plant is cheap to simulate, so sample efficiency is not the binding constraint, and because the per-agent credit of Section IV.B is a function of the current state and action that is recomputed exactly at every step rather than stored and replayed. Updates run on sequences: minibatches are segments of consecutive steps unrolled with gradient from a detached stored hidden state. Sampling shuffled individual timesteps, as we did initially, trains any recurrence as a one-step map and gives a recurrent encoder no temporal gradient at all [51]. A single illustrative run of `ippo_physics` kept under the old shuffled-timestep sampler reached $-0.213\,\text{s}^{-1}$, against -0.566 ± 0.123 for the same variant under the sequence-based sampler used everywhere else in this paper (Table 4); we report this as one run, not a claim, but it is the right order of magnitude for what a broken temporal gradient should cost. Algorithm 1 summarises the procedure.

D. Network architecture

The ablation in Section V.G names four architectural variants by the mechanism each adds; this section is what those names refer to, since a table of decay rates cannot substitute for knowing what `mocca` or `comm_raw` actually computes. Figure 3 shows the full pipeline, with the two optional stages — the encoder and the consensus rounds — and the optional action head shaded separately from the parts every variant shares.

Shared trunk Every variant feeds agent k's local observation $o_k^t \in \mathbb{R}^{12}$, concatenated with a one-hot agent identity and (when enabled) an incoming consensus message e_k^t, through a two-layer tanh multilayer perceptron of width 128 to a latent ℓ_k^t. Parameters are shared across agents; only the one-hot identity varies, which is what keeps the parameter count independent of the number of surfaces (hyperparameters tabulated in the Supplemental Material) and is what makes the eight-surface scaling probe in Section V.K possible without retraining a wider network.

Phasor encoder (`recurrent_only` and beyond) A single instantaneous accelerometer reading cannot distinguish a mode moving up from the same displacement moving down, so recovering phase needs memory. A per-agent GRU cell [51] carries a hidden state $h_k^t \in \mathbb{R}^{64}$ across the episode:

$$h_k^t = \mathrm{GRU}\Big(o_k^t, h_k^{t-1}\Big), \qquad \hat{\phi}_k^t = \tanh\big(W_{\mathrm{out}}\, h_k^t\big) \in \mathbb{R}^{2R}, \tag{5}$$

read as R=2 pairs $(\hat{\phi}_{k,r}^{\cos}, \hat{\phi}_{k,r}^{\sin})$, one per retained mode. The output is passed through tanh before it is used anywhere downstream: an unbounded phasor multiplied by a gain inside the action head saturates that head's own tanh at initialisation and kills the gradient before training starts, which is what the first implementation of this policy did. The phasor is carried as a $(\cos, \sin)$ pair rather than an angle throughout, so that wrap-around is not a discontinuity the network has to fight.

Phasor consensus (`mocca` and `comm_raw`) The communication mechanism is T=2 rounds of averaging over the spanwise line graph — agent k exchanges only with agents $k-1$ and $k+1$, which is what a distributed avionics bus physically offers. The operator

$$P = (1 - w)\, I + w\, A, \qquad w = 0.5, \tag{6}$$

with A the row-normalised adjacency of the line graph, is fixed rather than learned, so consensus adds no parameters; the bandwidth cost is the message itself, $2R$=4 numbers per agent per round regardless of how many surfaces there are. The consensus message is $e^t = P^T \hat{\phi}^t$ (T applications of P), read back into the trunk of each agent as e_k^t. `comm_raw` replaces $\hat{\phi}_k^t$ in this pipeline with the agent's raw local observation instead of the encoder's phasor estimate, testing whether the phasor abstraction itself is doing anything relative to simply broadcasting what a neighbour sees. When the health channel is enabled (used only in the residual-policy probes of Section V.K), each agent's own saturation fraction and its magnitude are concatenated onto the message before consensus, since phase alone cannot tell an agent that a neighbour's actuator has stopped responding.

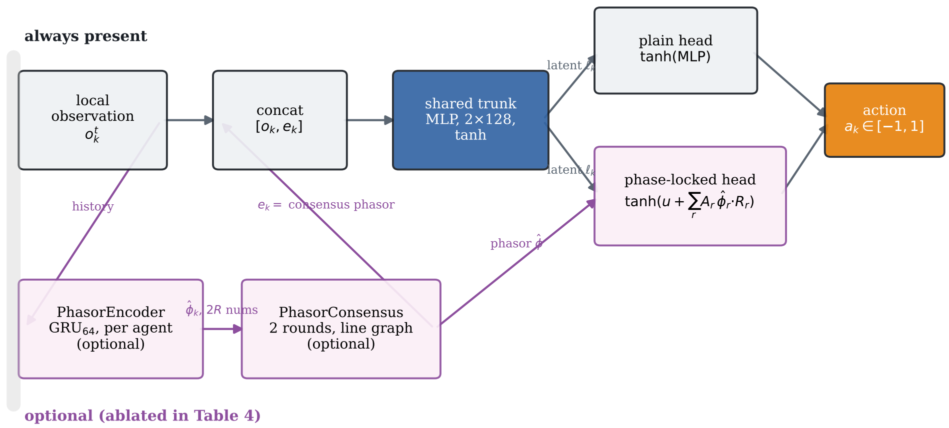

Fig. 3 The policy network. Grey boxes and the plain head are present in every variant; the pink boxes and the phase-locked head are the mechanisms ablated in Table 5, so `recurrent_only` keeps the encoder but not consensus or the head, `mocca_nohead` keeps the encoder and consensus but ends in the plain head, and `mocca` is the full diagram.

Phase-locked action head (the full `mocca` architecture) For retained mode r with consensus phasor $(\hat{\phi}_r^{\cos}, \hat{\phi}_r^{\sin})$, the head emits a gain $A_r \in [-1, 1]$ and a phase offset via a unit vector $(\cos\psi_r, \sin\psi_r)$, and acts by rotating the phasor:

$$\rho_r = \hat{\phi}_r^{\cos}\cos\psi_r - \hat{\phi}_r^{\sin}\sin\psi_r, \qquad a_k = \tanh\left(u + \sum_{r=1}^{R} A_r\,\rho_r\right), \tag{7}$$

where u is an additional broadband channel from the same latent. The resonant term is *added* to u rather than replacing it, which makes (7) a strict generalisation of the plain head: setting every A_r to zero recovers an unconstrained policy exactly. An earlier version of this head replaced u instead of adding to it, which turns a resonant prior into a restriction, and that version scored worse than removing the head entirely — the version reported as `mocca_nohead` in Table 5 is the trunk and consensus without this head, i.e. with the plain tanh(MLP) output shown at the top of Figure 3.

Critic and exploration The critic is a separate two-layer MLP that by default takes the same per-agent input as the actor; a fully centralised critic (fed the true modal state instead) was tried and made no measurable difference, consistent with this being a dense-reward, low-partial-observability credit problem rather than one where value estimation is the bottleneck, so we do not report it as a separate row. Every head, plain or phase-locked, has its final layer scaled by 0.01 at initialisation so a fresh policy is close to a no-op, which is what makes the residual variants of Section V.E well-posed from the first update rather than starting by fighting the base controller. Actions are Gaussian with a learned, state-independent log-standard-deviation shared across agents — the quantity swept in Section V.I.

V. Experimental evaluation

A. A baseline ladder that separates method from information

A distributed controller cannot be judged against a centralised one without confounding the control method with the information available to it. We report six rungs: collocated local rate feedback (no model, no communication); gain-scheduled LQR on the full 56-state plant including wake states, which is not implementable and serves as a ceiling; centralised LQG driven by a Kalman observer on the same six accelerometer channels the agents see; a fully decentralised LQG in which each surface runs its own observer on its own two accelerometers and commands only itself [25, 52]; and the same decentralised design re-tuned for control-authority margin (Section V.F). All are synthesised in discrete time at the control rate. The fully decentralised rung is the honest target.

B. Feasibility

We additionally run a fault-aware oracle: a gain bank re-synthesised per failure hypothesis, given the full state and told which surface failed. Its role is feasibility, not competition. Of nine evaluation conditions, 6 are stabilisable by this oracle; the rest are stabilisable by no controller with these actuators and are reported as infeasible rather than as failures of any method.

C. Metrics

PPO return cannot rank runs whose reward functions differ, so every controller is scored on the same physical quantities over identical episodes — same airspeeds, initial deformation, turbulence and injected failure: the fraction of episodes leaving the linear regime, the decay rate (1), peak tip plunge, and RMS deflection. Following [39] we report effect sizes and significance rather than means alone.

D. Training convergence

Figure 4 gives the training-time behaviour behind every learned number in this section: mean structural energy and divergence rate over training on the curriculum task distribution, averaged across seeds. Every variant is within a small margin of its asymptote by 750 000 of the 1 500 000 environment steps budgeted, confirming the training budget was chosen with headroom rather than tuned to convergence after the fact. `residual_only` starts substantially lower than every other variant and stays there for the first 200 000 steps, the training-time signature of the small-gain initialisation (Section IV.D): it inherits the decentralised LQG base law's performance from update one rather than the near-open-loop performance every other variant starts from. The divergence rate — a training statistic, not the evaluation metric of Table 3 — spikes above 0.5 % only in the first few thousand steps, before any policy has learned anything and while the curriculum is still easing in (Algorithm 1), and settles below 0.15 % thereafter for every variant: the curriculum keeps training itself stable even where the resulting policy later struggles on the harder end of the evaluation grid.

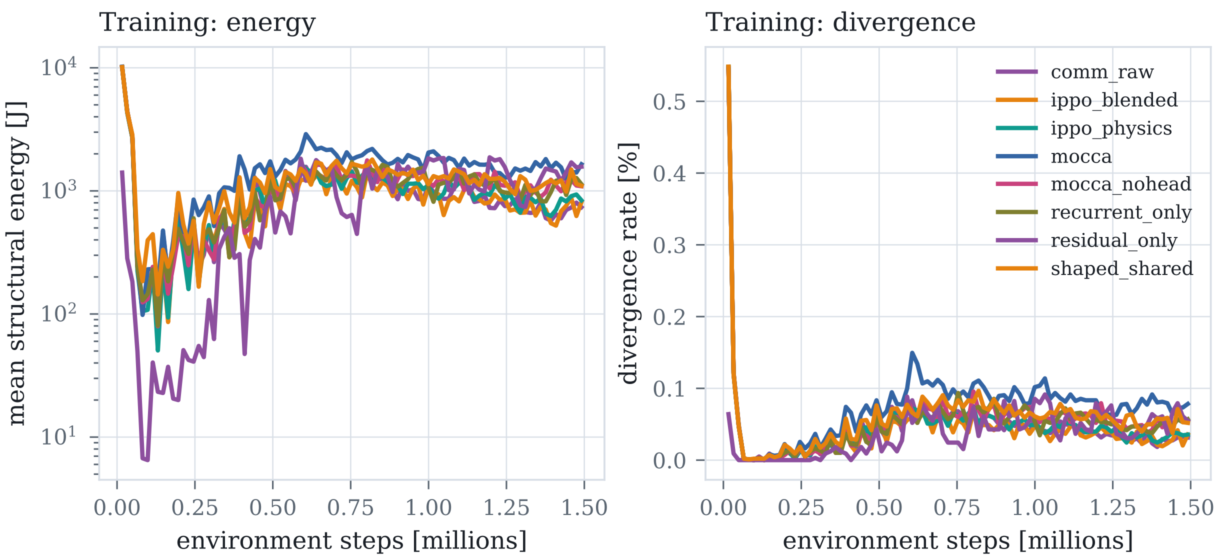

Fig. 4 Training-time mean structural energy (left, log scale) and divergence rate (right) against environment steps, averaged over seeds, for every credit and architecture variant. All variants reach a stable plateau well inside the training budget; `residual_only` converges fastest because it starts from a decentralised LQG base law rather than from scratch.

E. Results

Before the aggregate numbers, Figure 5 shows what a single trial looks like: the same 5 cm pluck at $U/U_\mathrm{f} = 1.10$ under open loop, local rate feedback, and centralised LQR, sharing axes. Open loop grows into the divergence stop within 0.93 s; local rate feedback holds a bounded, undamped oscillation rather than suppressing it, the signature of a controller that reacts to local motion without engaging the coupled flutter mode; centralised LQR removes six orders of magnitude of structural energy within 1.5 s. Every entry in Table 3 is a decay rate fitted to a trace of this shape, at one of nine flight and failure conditions.

Three things in Table 3 matter.

First, the classical ladder separates by information structure rather than sophistication. Centralised LQG reaches $-5.52\,\mathrm{s}^{-1}$ on healthy conditions; the fully decentralised LQG, seeing exactly what the agents see, reaches -2.05 — a factor of 2.7 cost from decentralisation, consistent with the classical expectation [25, 27], and why the decentralised controller is the reference for a distributed policy.

Second, the best learned policy reaches $-1.53\,\mathrm{s}^{-1}$, about three quarters of the decentralised classical result. The average conceals more: broken out by condition the decentralised LQG achieves -5.42, -1.07 and $+0.32$ at 1.05, 1.25 and 1.45 U_f — failing outright at the top of the envelope — against -1.98, -1.92 and -0.69 for the learned policy, which is far more uniform across the envelope and holds a condition where the classical one diverges, while losing badly where the classical one is strongest.

Third, and least comfortable: a robustness advantage measured at a conventional exploration scale does not survive

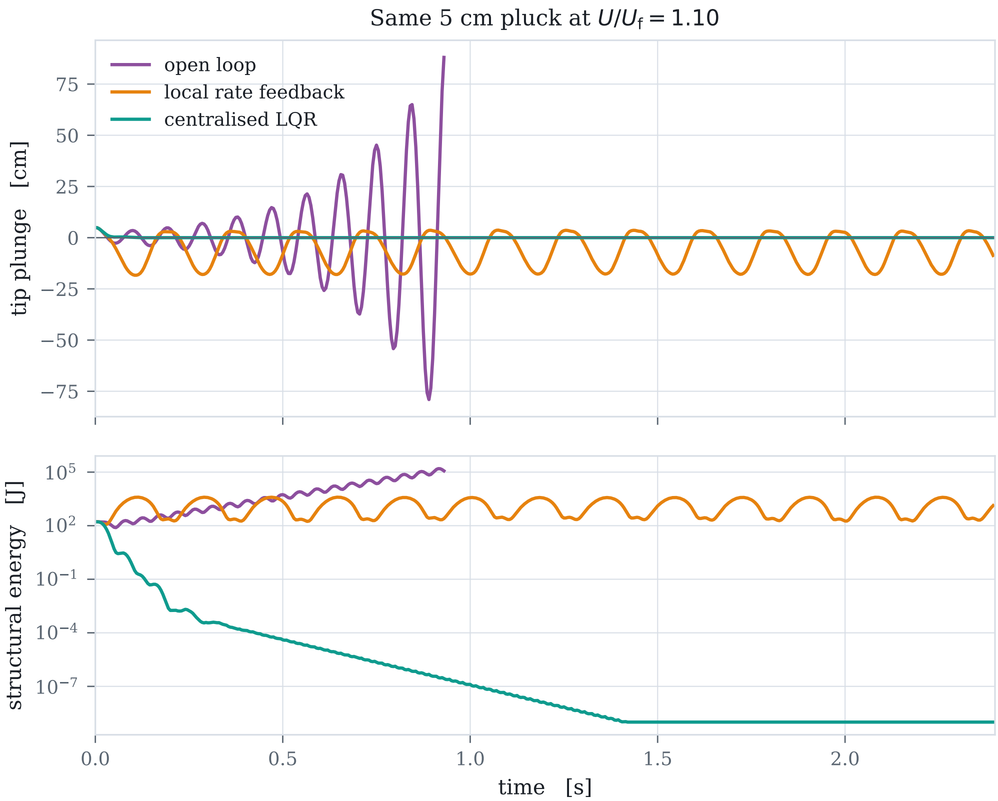

Fig. 5 Tip plunge (top) and structural energy (bottom, log scale) for the same initial pluck under three of the five classical rungs. Local rate feedback neither grows nor decays: it damps local motion but never engages the unstable mode, which Section V.H confirms directly.

at the scale that maximises damping. At $\sigma = 0.30$ the learned policies held all six feasible conditions, matching the fault-aware oracle; at $\sigma = 0.05$ they hold five, as the classical controllers do, while damping three times better. Exploration scale trades robustness against nominal performance; we report both halves.

F. Is the classical baseline's fragility at the envelope edge a tuning artefact?

The comparison above rests on `dlqg_local` being a fair classical target, and an unstructured LQG has no guaranteed stability margin, unlike full-state LQR [53], so we checked whether a better-tuned classical design closes the gap rather than assuming either way.

Loop transfer recovery [54] — the standard remedy for an *observer-induced* margin loss — does not help: boosting the Kalman filter's process-noise weighting toward the full-state loop shape across four orders of magnitude leaves the closed-loop control-authority margin at 1.45 U_f unchanged (0.158 throughout; Supplemental Material, Section S4), because the fully state-fed loop already sits at the same margin — the observer is not where the fragility lives. Re-weighting the state-feedback gain itself does move it: a 50× increase in the control-effort weight raises the margin from 0.158 to 0.179 before plateauing, at a measurable cost to nominal decay rate (the closed-loop growth rate at design

Table 3 Energy decay rate by controller and flight condition. Negative is suppressed, `D` marks the fraction of episodes that diverged, and `–` marks conditions infeasible for any controller.

Condition	local_fb	lqr_fixed	lqr_sched	lqg_local	dlqg_local	dlqg_robust	lqr_oracle	comm_raw	ippo_blended	ippo_physics	mocca_nohead	mocca	recurrent_only	residual_only	shaped_shared
U=1.05Uf healthy	-0.06	-4.40	-4.61	-4.71	-5.42	-4.17	-4.54	-0.78	-0.99	-0.90	-0.68	-0.85	-0.82	-1.98	-0.96
U=1.05Uf jam#2@8deg	-0.10	-0.10	-0.12	-0.13	-0.09	-0.11	-0.06	-0.05	-0.06	-0.06	-0.08	0.01	-0.08	-0.07	-0.10
U=1.05Uf jam#2@14deg	-0.12	-0.09	-0.10	-0.11	-0.08	-0.10	-0.06	-0.06	-0.08	-0.07	-0.06	D42%	-0.08	-0.07	D33%
U=1.25Uf healthy	-0.01	-5.27	-4.93	-5.38	-1.07	-0.82	-4.93	-0.62	-0.96	-0.77	-0.46	-0.47	-0.51	-1.92	-0.65
U=1.25Uf jam#2@8deg	D100%	D100%	D50%	D100%	D50%	D50%	-0.12	D75%	D100%	D100%	D100%	D100%	D100%	D50%	D100%
U=1.25Uf jam#2@14deg	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U=1.45Uf healthy	D100%	-5.79	-5.68	-6.45	0.32	0.18	-5.74	-0.39	-1.30	-0.03	0.21	0.15	-0.09	-0.69	-0.47
U=1.45Uf jam#2@8deg	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U=1.45Uf jam#2@14deg	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Stabilised	4/6	5/6	5/6	5/6	5/6	5/6	6/6	5/6	5/6	5/6	5/6	4/6	5/6	5/6	4/6

conditions worsens from 0.9744 to 0.9916 in discrete-time terms). That a fifty-fold retuning buys back so little margin, at a real performance cost, is itself the answer: the fragility at $1.45\,U_\mathrm{f}$ is structural to decentralised control at this operating point, not a tuning artefact.

`dlqg_robust` in Table 3 is this re-tuned design, carried through the full nonlinear evaluation: it improves the $1.45\,U_\mathrm{f}$ healthy-condition growth rate from +0.32 to +0.18 s^{-1} — smaller, but still growing — while giving up nominal performance elsewhere (-0.82 against $-1.07\ \mathrm{s}^{-1}$ at $1.25\,U_\mathrm{f}$). Neither classical design holds the condition the learned policy holds. The margin found above (0.158–0.179, stable down to 15.8 % to 17.9 % of design authority) is also comfortably clear of the plant's own measured control-authority discrepancy relative to UVLM (a 1.9× overestimate, Section III.F, true authority at $1/1.9 \approx 53$ % of design), a safety factor of roughly 3× in relative-authority terms, in the direction that already occurred: the classical baseline's fragility at the envelope edge is not explained away by either an under-tuned weight or the plant's own modelling error.

G. Ablations

Credit assignment Six seeds per variant at $\sigma = 0.05$ (Table 4), five variants: the exact-plus-shaping combination (`ippo_blended`), each half alone (`ippo_physics`, `shaped_shared`), the naive undecomposed shared reward (`ippo_shared`), and a learned value-decomposition credit in the style of VDN [9] (`vdn_shared`, Section V.G). Five variants means ten pairwise comparisons rather than three, so every p-value here and in Table 4 is Holm-Bonferroni-corrected across that full family, with effect sizes as Hedges' g (small-sample-corrected) alongside Cohen's d. A preliminary, lower-seed sweep of the credit-versus-shaping comparison read positive, then null, then positive again as seeds were added, before six seeds stabilised its sign, which is why we adopted a fixed seed count and significance test rather than reading conclusions off small samples; that history does not invalidate the six-seed measurement, but is part of why we treat these comparisons as being at the edge of what six seeds can settle.

The combination reaches -1.085 ± 0.243, and beats the closed-form credit alone (-0.566 ± 0.123) at $g = -2.49$,

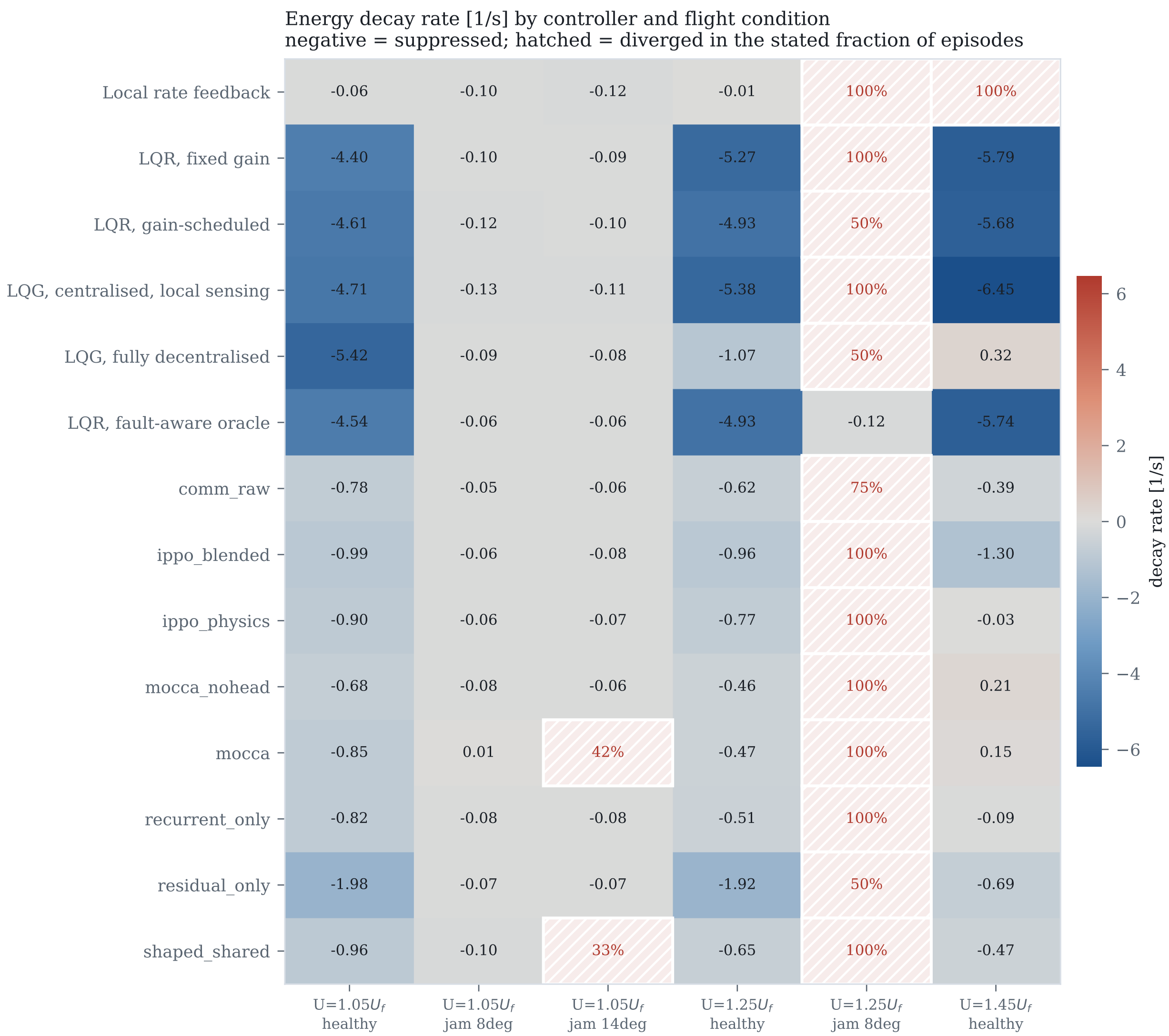

Fig. 6 Decay rate by controller and condition. Hatching marks divergence.

$p_{\mathrm{Holm}} = 0.020$ – the one comparison in this table that survives correction. It does *not* significantly beat shaping alone (-0.690 ± 0.145, $g = -1.82$, $p_{\mathrm{Holm}} = 0.080$) at this standard: tested alongside only two other variants, that comparison's uncorrected $p = 0.009$ looks significant, but it does not survive being tested as part of the five-variant family the physical question actually calls for. We do not think the mechanism we described – the closed form supplying an instantaneous term, the shaping supplying a long-horizon one – is wrong, only that this experiment does not establish it at the standard we now hold every comparison in this table to; a seed count larger than six, which is what the variance at six seeds leaves headroom for, is what would settle it.

The naive shared reward is not the outright failure a single seed originally suggested: at six seeds it averages -0.526 ± 0.806, damping well on some seeds (-1.969) and slightly exciting on others ($+0.060$), a spread three to six times wider than any credit-bearing variant's and indistinguishable from all three after correction ($p_{\mathrm{Holm}} \geq 0.57$). It

Table 4 Credit-assignment ablation at $\sigma = 0.05$, six seeds per variant, five variants: the closed-form credit, log-energy shaping, their combination, a naive undecomposed shared reward, and a learned value-decomposition credit (Section V.G). All ten pairwise p-values are Holm-Bonferroni corrected across this family.

Variant	Seeds	Decay rate [1/s]	Conditions held
`ippo_blended`	6	-1.085 ± 0.243	5.0/9
`ippo_physics`	6	-0.566 ± 0.123	5.0/9
`ippo_shared`	6	-0.526 ± 0.806	4.0/9
`shaped_shared`	6	-0.690 ± 0.145	4.3/9
`vdn_shared`	6	-1.670 ± 0.699	1.8/9

Comparison		Cohen's d	Hedges' g	p	p_{Holm}
`ippo_blended`	vs `ippo_physics`	−2.69	−2.49	0.0020	0.0198*
`ippo_blended`	vs `ippo_shared`	−0.94	−0.87	0.1557	0.5683
`ippo_blended`	vs `shaped_shared`	−1.97	−1.82	0.0088	0.0795
`ippo_blended`	vs `vdn_shared`	1.12	1.03	0.0996	0.4980
`ippo_physics`	vs `ippo_shared`	−0.07	−0.06	0.9078	1.0000
`ippo_physics`	vs `shaped_shared`	0.92	0.85	0.1421	0.5683
`ippo_physics`	vs `vdn_shared`	2.20	2.03	0.0112	0.0895
`ippo_shared`	vs `shaped_shared`	0.28	0.26	0.6426	1.0000
`ippo_shared`	vs `vdn_shared`	1.52	1.40	0.0257	0.1543
`shaped_shared`	vs `vdn_shared`	1.94	1.79	0.0177	0.1240

is not that it fails, but that it is far less reliable: run-to-run variance several times that of the credit-bearing variants spans the range from competitive to mildly destabilising. The value-decomposition credit is the opposite failure mode: its mean, -1.670 ± 0.699, is the most negative of all five and beats the closed-form credit alone at $g = -2.03$, $p_{\mathrm{Holm}} = 0.090$ — short of significance but the largest point estimate in the table — while its variance is nearly three times `ippo_blended`'s, and it stabilises only 1.8 of nine conditions on average against 5.0 for every credit-bearing variant: it damps harder where it stabilises at all, at the cost of stabilising far fewer conditions. These are the two failure modes a learned credit signal can have when the environment does not hand it one: no decomposition gives high-variance, unreliable performance; a decomposition estimated by summing per-agent value heads to a team target gives aggressive but inconsistent performance. The physics-derived signal is the only one in this comparison whose seed-to-seed spread (0.123–0.243) and stabilised-condition count (5.0/9, matched by `ippo_physics` and `ippo_blended`) are both consistent across seeds, a property worth weighing alongside the mean.

A learned value-decomposition credit baseline. VDN [9] and QMIX [10] solve a value-based, discrete-action credit-assignment problem: a joint action-value function is decomposed into per-agent utilities so the greedy action for each agent can be read off independently. Our setting is continuous-action, on-policy actor-critic, so we adapt the decomposition: each agent keeps its own decentralised state-value head $V_i(o_i)$, but under the naive shared reward the summed value $V_{\mathrm{tot}} = \sum_i V_i$ is fit to the team return via generalised advantage estimation [48] computed once at the team level, and every agent's policy update uses that shared advantage. This decomposes the *value function*, as VDN's architecture does, without reproducing VDN's original mechanism, which relies on each agent's utility depending on its own discrete action. The result: better raw damping where it holds a condition, worse and more variable robustness overall than either the naive baseline or the closed-form credit. We read this as evidence that VDN-style decomposition, adapted this way, is not a free improvement over an undecomposed shared reward here, not that value decomposition cannot work in some other form; COMA's counterfactual baseline [11] and QMIX's monotonic mixing network [10] were not adapted and tested, so this opens the comparison rather than closing it.

Architecture The mechanisms we derived from the same physics do not survive measurement, and at six seeds per variant none of the four survives even marginally (Table 5). A plain feedforward policy with the blended credit reaches -1.085 ± 0.243; adding a recurrent phasor encoder gives -0.474 ± 0.249, adding phasor consensus -0.308 ± 0.169, adding the phase-locked action head (the full architecture) -0.389 ± 0.219, and sharing raw neighbour observations -0.596 ± 0.227. All four are significantly worse than the plain policy after Holm-Bonferroni correction across the ten pairwise comparisons among the five variants in this family (the four architecture variants plus the plain `ippo_blended` policy they are each measured against), reporting Hedges' g alongside Cohen's d to correct for the small-sample bias at $n = 6$ ($g = -2.29$, $p_{\text{Holm}} = 0.013$ for the recurrent encoder; $g = -3.43$, $p_{\text{Holm}} = 0.001$ for consensus; $g = -2.78$, $p_{\text{Holm}} = 0.004$ for the full architecture; $g = +1.92$, $p_{\text{Holm}} = 0.034$ for raw neighbour sharing). Unlike the credit comparison below, every one of these four survives the stricter accounting. Every elaboration we proposed made things worse, and correcting for the number of comparisons made the conclusion no less sharp.

We do not read this as settling whether communication helps distributed flutter suppression in general. Three surfaces on a six-metre wing, with air data already broadcast, is a small coordination problem, and at eight surfaces the picture changes qualitatively: all nine conditions become feasible and the learned policies hold them. It does settle that on this problem the credit signal was worth deriving from the physics and the message content was not.

Figure 7 puts every controller from Sections V.E and V.G on common axes. The pattern is visible at a glance in a way the tables cannot show as directly: the learned variants cluster with the classical rungs on the robustness axis (top) while forming a visibly separate, weaker band on the nominal-performance axis (bottom) — with one exception. The residual policy, which corrects a decentralised LQG rather than replacing it, is the only learned variant that reaches into the range the classical controllers occupy on *both* axes at once.

H. Emergent spanwise phase coordination

"Emergent coordination" is easy to assert and hard to pin down, so we define it as something measurable: the phase at which each surface deflects relative to the critical structural mode's velocity, as a function of spanwise station. A controller with no notion of the mode shape should show no systematic phase progression along the span; one that has discovered the mode shape should show a consistent progression matched to it. The measurement — the cross-spectrum

Table 5 Architecture ablation at $\sigma = 0.05$. Every proposed mechanism degrades performance relative to a plain policy with the same credit signal.

Variant	Seeds	Decay rate [1/s]	Conditions held
`comm_raw`	6	-0.596 ± 0.227	5.0/9
`mocca`	6	-0.389 ± 0.219	4.2/9
`mocca_nohead`	6	-0.308 ± 0.169	5.0/9
`recurrent_only`	6	-0.474 ± 0.249	5.0/9

Comparison		Cohen's d	Hedges' g	p	p_{Holm}
`comm_raw`	vs `mocca`	−0.93	−0.86	0.1391	0.6953
`comm_raw`	vs `mocca_nohead`	−1.44	−1.33	0.0336	0.2018
`comm_raw`	vs `recurrent_only`	−0.51	−0.47	0.3961	1.0000
`mocca`	vs `mocca_nohead`	−0.41	−0.38	0.4907	1.0000
`mocca`	vs `recurrent_only`	0.36	0.33	0.5444	1.0000
`mocca_nohead`	vs `recurrent_only`	0.78	0.72	0.2104	0.8414

of each surface's deflection against the first bending mode's velocity, at the frequency where the modal response peaks — is applied identically to every controller, so a learned pattern can be held against what optimal centralised synthesis produces.

Applied to the classical ladder the instrument already separates the rungs sharply (Figure 8, first three series). Local rate feedback is essentially flat at $-15°$ per semispan and responds at 5.50 Hz: it never engages the unstable mode at all. The fully decentralised LQG reaches $+27°$ per semispan at 7.50 Hz, and centralised LQG develops $+90°$ per semispan at 11.00 Hz — matching the 11.03 Hz the eigenvalue analysis of Section III.F's plant predicts almost exactly. The pattern is monotone: the more capable the controller, the stronger its spanwise phase progression, and only the optimal centralised design locks onto the true flutter frequency.

The learned policies give this a genuine test. The strongest credit-only policy, `ippo_blended`, locks onto 11.00 Hz — the same frequency as centralised LQG and the eigenvalue prediction — with a phase gradient of $+121°$ per semispan, steeper than the centralised design's, reached from six accelerometer channels with no communication and no information about the mode shape beyond what the reward and dynamics implicitly encode. The residual policy locks onto a different, higher-frequency mode ($+170°$ per semispan at 15.50 Hz) rather than the primary flutter mode, consistent with it correcting a base law that already suppresses the primary mode. We read the first result as the concrete, falsifiable content behind "emergent" in this paper: a distributed policy that has never been told the mode shape converges, from local measurement alone, on the same resonant frequency that full-state, full-model synthesis converges on by construction.

The same measurement corrects a wrong inference authority alone would suggest. Local rate feedback uses 44.7 % of available deflection to reach $-0.03\,\mathrm{s}^{-1}$; centralised LQG uses 0.7 % to reach -5.52. The better the controller, the *less* authority it spends, because flutter suppression is a phase-precision problem rather than a control-power one. The

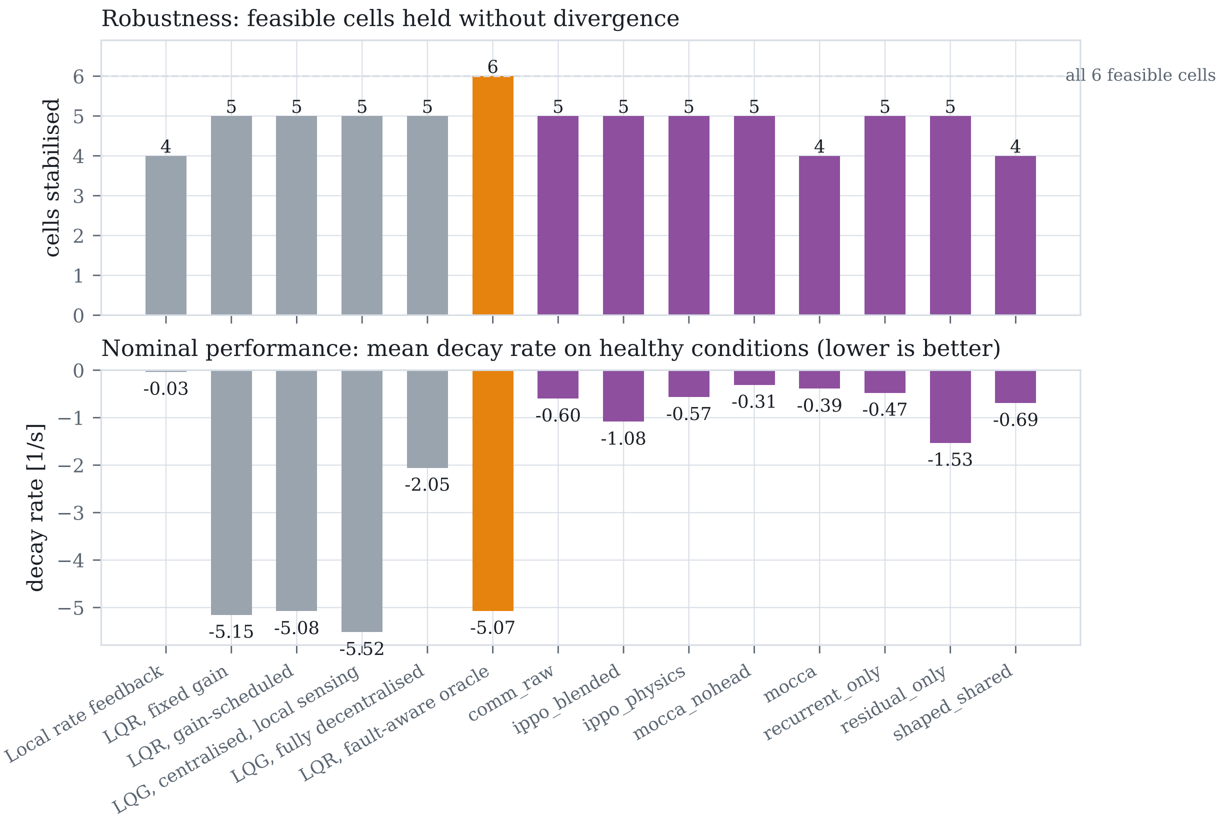

Fig. 7 Every controller and variant on one axis. Top: feasible conditions held without divergence, out of the six the fault-aware oracle shows are stabilisable. Bottom: mean decay rate on healthy conditions. The learned variants (purple) match the classical rungs on robustness while trailing them on nominal damping, except for `residual_only`, which is the strongest learned result on both axes simultaneously.

learned policies (1.2 % to 1.3 % authority) are accordingly not under-actuated; they apply authority comparable to the good classical controllers, at a phase closer to optimal than local feedback but not as sharp as the centralised design.

I. What limits learning here

Figure 9 is the finding with the widest reach, and we hold it to the same six-seed standard as the ablations above rather than to a smaller pilot sweep. Holding everything else fixed and varying only the initial exploration standard deviation moves performance by a factor of two (-1.805 ± 0.257 at $\sigma = 0.05$ against -0.908 ± 0.271 at $\sigma = 0.15$, a ratio of 1.99), with an interior optimum at $\sigma = 0.05$ and a clear degradation above it – the qualitative shape an earlier, thinner sweep already suggested, and which six seeds confirms rather than overturns.

The mechanism can be read off the axis. A controller achieving the classical result uses about 0.105° RMS of deflection; $\sigma = 0.30$ corresponds to 4.5°. The policy's own exploration is then some forty times the amplitude that works, and no credit decomposition can recover a signal buried underneath it.

This is not a tuning remark. Before we found it, every variant plateaued between -0.03 and $-0.82\,\mathrm{s}^{-1}$ regardless of

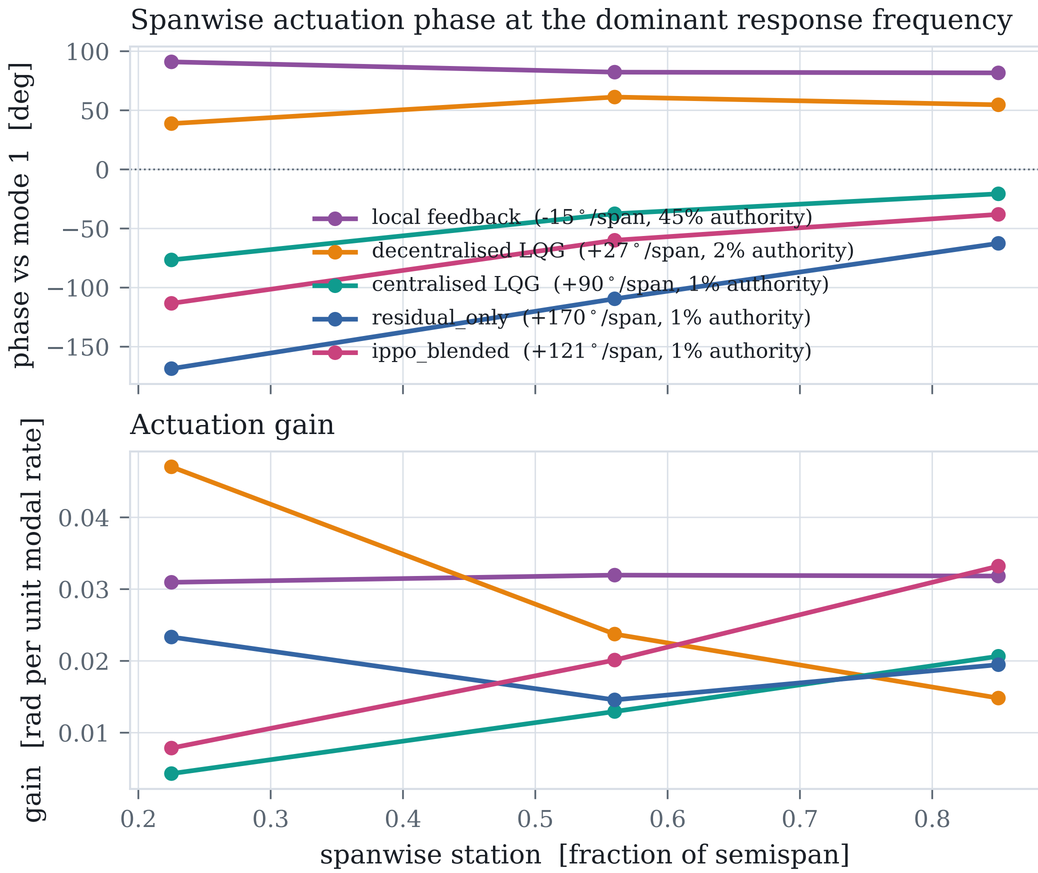

Fig. 8 Spanwise actuation phase (top) and gain (bottom) at each controller's dominant response frequency. A sloped phase profile is a travelling-wave actuation pattern matched to the mode shape; a flat profile is purely local reaction. Legend entries give the phase gradient and the RMS actuation authority used.

credit signal, communication, action parameterisation, memory, auxiliary supervision or base controller. An initial two-seed pilot of the spread behind that plateau suggested it fell sharply between the conventional and matched scale (0.132 at $\sigma = 0.30$ to 0.016 at $\sigma = 0.05$); the six-seed measurement reported here does not confirm that (0.325 at $\sigma = 0.30$ to 0.257 at $\sigma = 0.05$), a second instance in this paper of a two-seed spread understating the true variance, alongside the credit-versus-shaping correction in Section V.G. The interior optimum and the factor-of-two effect on the mean are what survive at six seeds; the claim that the spread itself shrinks sharply at the matched scale does not, and we do not make it. A capability probe rules out the competing explanation: training on the single easiest condition gives +0.21 and +0.27, worse than training on the full distribution, so the plateau was never a matter of the task being too broad.

The practical recommendation is specific. The useful actuation amplitude is computable before any training, from a classical design or from an estimate of the damping authority required. Scale the exploration to it. On this problem that single number mattered more than every algorithmic choice we made, and it is cheap to check.

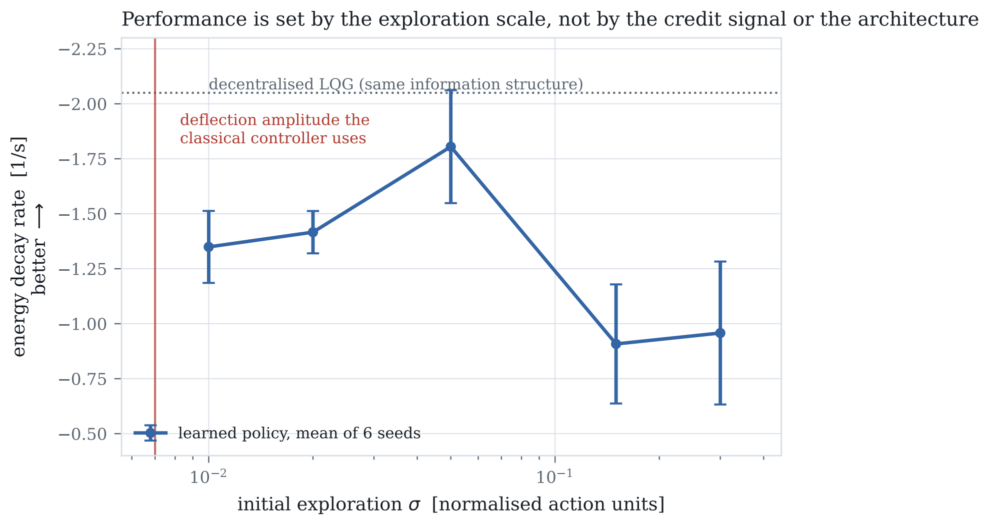

Fig. 9 Decay rate against initial exploration standard deviation, all else fixed. The vertical line marks the deflection amplitude the classical controller uses. Six seeds per point.

J. Sensitivity to control-influence identification error

The credit signal's sign is decided by the modal residue $v_r^\top b_k$ (Section IV), free in simulation but requiring an identified control-influence estimate on hardware. A wrong sign rewards a surface for exciting a mode rather than damping it, so we quantify how much identification error it tolerates.

Analytic: how large an error flips a sign Modelling identification error as an independent relative perturbation on each $(mode, surface)$ entry of B, scaled to that mode's own row, the most fragile entry — the mid-span surface's (`aileron_mid`) residue on the first torsion mode — flips sign at just 11 % relative error, essentially speed-invariant across the flight envelope (11.0 % to 11.0 % at 1.05–1.45 U_f). Every other entry is more robust, up to entries that do not flip within ±100 % error at all: for the least robust surface–mode pairing this is a real vulnerability, at an error smaller than many real aerodynamic-derivative uncertainties; for the rest, not.

Under a *uniform* per-surface scale error instead — every entry of a surface's column moving by the same fraction, as a simple overall effectiveness mis-estimate would — every sign flips simultaneously at exactly −100 % relative error (classical control reversal). The plant's own reported UVLM control-authority discrepancy (Section III.F) is +90 %, an overestimate that moves the assumed value *away* from zero: a safety factor of roughly 2–3× in relative-error terms, in the direction that already occurred.

Empirical: training with a misinformed credit signal We retrained the exact-plus-shaping credit variant with the true dynamics held fixed and only the credit signal's belief about control authority perturbed by a uniform ±60 %, three seeds each. At the unperturbed baseline the healthy decay rate is −0.441 ± 0.330; believing 60 % more authority gives

-0.259 ± 0.217, and 60 % less gives -0.570 ± 0.131. Neither direction collapses performance or reverses sign, though the spread across all three conditions at three seeds is wide enough that we do not read a monotone trend into the ordering. Both perturbations stay well short of the -100 % sign-reversal boundary the uniform-scale analysis above predicts, so this is evidence that moderate uniform identification error degrades gracefully rather than catastrophically, not a test of what happens past that boundary.

K. Additional probes

Three further questions arising directly from the results above were checked at one or two seeds each rather than the six-seed main standard, and are reported as probes rather than findings in the Supplemental Material: whether the architecture ablation's negative communication result survives moving to a residual policy (it does), whether supervising the phasor encoder on the true modal phasor helps (it does not), and what eight-surface scaling changes beyond the feasible-condition count reported above (mainly that "held" there means bounded rather than decisively damped). None of the qualitative claims made elsewhere in this paper rest on these probes.

VI. Discussion and conclusion

A. Discussion

Pulling the pieces together: this study set out to ask whether treating each control surface as a separate learning agent is a workable way to suppress flutter, and the answer is conditional. On credit assignment the answer is yes, with numbers behind it; on whether the resulting policies are ready to fly, not yet, and the gap to a classical decentralised design is real rather than an artefact of tuning (Section V.E).

For practitioners The most actionable finding is the least glamorous one. Before training a policy for a physical control task, compute the actuation amplitude a competent classical controller would use, and check the initial exploration scale against it. On this plant the two were separated by roughly a factor of forty at a conventional PPO setting, and every algorithmic refinement we tried was invisible until that gap was closed. We expect this check to be worth running on any task where a classical baseline is available even loosely.

Toward a certifiable architecture Flutter suppression is flight-safety-critical by construction (Section I): failure is loss of the aircraft, with no time for a pilot to intervene. Nothing in this paper argues that the learned policies as trained are ready for that role. Every classical rung has a synthesis procedure with a known stability margin (Supplemental Material, Section S4); a PPO-trained network has neither a certified margin nor a certification pathway, and we are not aware of one for any learned flutter-suppression policy in the literature we survey, our own included. We place this work at approximately TRL 2–3 (technology concept formulated, analytical proof of concept): a validated simulation testbed and a credit signal derived from first principles, with no wind-tunnel or hardware-in-the-loop evidence yet.

The residual formulation offers the most concrete path we see toward certifiable. `residual_only` (Section V.E) is, by construction, a decentralised LQG base law — itself certifiable by the classical route — plus a learned correction initialised near zero and scaled by a fixed authority fraction. A run-time-assurance architecture on the same structure — the certified base law as the default, the learned correction bounded to a fixed authority fraction, and a monitor that reverts to the base law if the plant's state leaves a pre-verified envelope — would let the learned component fail safe into a controller already certifiable on its own. We have not built or verified such a monitor, nor characterised how the credit signal's identification-error robustness (Section V.J) interacts with the monitor's authority bound; both are the concrete next step this result points to.

For the credit-assignment literature The result in Section V.G is a genuine instance of a difference reward with no estimation step at all, on a task with real physical structure rather than a toy coordination game, and it beats the closed-form credit used alone at the standard we hold every comparison in this paper to. Against shaping used alone, a smaller three-variant, uncorrected comparison would look significant ($p = 0.009$); once compared as part of the five-variant family the physical question calls for, it does not survive correction, and we report the null result rather than the more favourable uncorrected one. Against an adapted value-decomposition baseline (Section V.G), the closed-form credit's advantage is reliability rather than raw mean: the learned decomposition damps harder where it stabilises, at the cost of stabilising fewer conditions and roughly triple the seed-to-seed variance. The physics-derived signal is competitive with, not decisively better than, the alternatives we tested, and its main advantage is that it requires no estimation step and stays consistent across seeds and conditions in a way none of the estimated alternatives were.

On the negative architecture result We did not expect the phasor consensus and phase-locked action head to hurt. A plausible interpretation is that a three-surface, six-metre wing with broadcast air data is simply too small a coordination problem for an explicit channel to earn its bandwidth cost, and the eight-surface result — where every condition becomes feasible, albeit with weak nominal damping (Section V.K) — is at least consistent with that story, though not run to the statistical standard we hold the main claims to. We read this as a caution against building communication machinery into a distributed controller by default, on the strength of a physical argument, without first checking whether a plain policy with the same credit signal already does the job.

Why we still find phase coordination the most striking result A policy trained end to end on a shaped scalar reward, using only its own two accelerometers, converges on the same resonant frequency that a full-state Kalman-filtered LQG converges on by construction (Section V.H). That frequency is not supplied anywhere in the reward, the observation, or the network architecture; it is a property of the coupled structure that the policy has to find. We think this is the sense in which the word "emergent" is earned here, as distinct from being merely asserted.

B. Limitations

- The learned policies do not match the decentralised classical controller on average nominal damping (-1.53 against $-2.05\,\mathrm{s}^{-1}$). They are more uniform across the envelope and hold a condition where it diverges; we do not claim they dominate it. That the classical rung is genuinely, not just weakly-tuned, fragile at that condition is itself checked in Section V.F rather than assumed.
- The robustness result is scale-dependent, as reported in Section V.E. We give the trade-off rather than its better half.
- The plant over-estimates control authority by a factor of 1.9 relative to UVLM (Section III.F). Relative results are unaffected; absolute decay rates would not transfer. We have not retrained on the UVLM plant. The same simplification (independent, per-surface flap strips with no spanwise aerodynamic carryover between them) is what makes the credit decomposition of Proposition 1 exactly additive; we have not characterised how the decomposition would degrade under a higher-fidelity aerodynamic model with inter-surface coupling, only how much control-authority error the classical baseline's stability tolerates (Section V.F) and how much identification error the credit signal's sign tolerates (Section V.J).
- The credit signal requires the modal control influence $v_r^\top b_k$, free in simulation but requiring a modal model on hardware. Its sensitivity to identification error is characterised in Section V.J, not asserted away: the most fragile surface–mode pairing tolerates only 11 % independent relative error before its sign reverses, though the plant's own measured discrepancy is well clear of that boundary.
- Proposition 1 concerns the difference reward, not optimality of the resulting policy.
- Ablations use six seeds for the credit comparison (now five variants, Section V.G), six for the architecture comparison, and now six for the exploration-scale sweep (Section V.I), all on a three-surface wing. The eight-surface results, and the three further questions in Section V.K, are one- and two-seed probes and are not run to the same standard. Correcting for the number of pairwise comparisons made across the five-variant credit family (Holm-Bonferroni) changed which comparisons are significant, and bringing the exploration sweep to six seeds changed our own estimate of its seed-to-seed spread substantially, though not the qualitative shape of the curve; see Sections V.G and V.I for what did and did not survive.
- The exploration finding is established on this plant and this algorithm. We expect it to generalise to control tasks whose useful actuation band is narrow relative to the action range, but have not shown that.
- The learned policies have no certification pathway and no stability-margin guarantee, unlike every classical rung in the baseline ladder. We place this work at approximately TRL 2–3 and describe what we think the concrete next step would be, without having taken it, in Section VI.

C. Conclusion

Posing flutter suppression as a distributed learning problem exposes a credit structure the physics resolves in closed form: the difference reward is an agent's own control power, normalised by the instantaneous energy so that it coincides with the objective and cannot be gamed by sustaining the oscillation. Combining it with log-energy shaping beats the closed form used alone once compared, and corrected for multiple comparisons, against the full family of alternatives a reviewer would want to see it measured against; against estimated credit signals, its advantage is reliability across seeds and conditions rather than a uniformly larger effect.

The more transferable lesson is the caution. That result, and a clean negative result on the communication machinery, were both invisible until the exploration scale was matched to the actuation amplitude that damps the wing — a quantity computable in advance from a classical design. On this problem it mattered more than any algorithmic choice we made, and we suspect that is true of active vibration control more broadly.

A. Testbed parameters

Table 6 gives the geometric and structural parameters of the wing and Table 7 the actuator and sensing parameters, common to every controller in this study. All are the values used to generate Table 2.

Table 6 Wing geometry and structural properties.

Parameter	Value
Semispan	6.096 m
Chord	1.8288 m
Mass per unit length	35.71 kg m^{-1}
Polar inertia per unit length (about EA)	8.64 kg m
Bending stiffness EI	9.77×10^6 N m^2
Torsional stiffness GJ	0.987×10^6 N m^2
Elastic axis	33 % chord
Centre of gravity	43 % chord
Bending / torsion modes retained	2 / 2
Aerodynamic strips	24
Air density (sea level)	1.225 kg m^{-3}

B. Training hyperparameters, compute, and reproducibility

PPO and network hyperparameters, per-run wall-clock and GPU-time totals, and the software/hardware stack are tabulated in the Supplemental Material (Sections S1-S2) rather than here, to stay within the journal's length guidance; every learned variant in this paper trained on a single workstation GPU, and no result required infrastructure beyond that.

Table 7 Control surface, actuator and sensing parameters (identical for all three surfaces).

Parameter	Value
Spanwise bands (fraction of semispan)	$[0.05, 0.40]$, $[0.40, 0.72]$, $[0.72, 0.98]$
Hinge location	75 % chord
Deflection travel	$\pm 15°$
Rate limit	$200\,°\,\mathrm{s}^{-1}$
Actuator time constant	20 ms
Control update rate	200 Hz
Local observation per agent	2 accelerometers, plunge/twist rate, own saturation, air data

C. Baseline controller synthesis

The synthesis equations behind every rung of the classical baseline ladder — the augmented discretised plant, the LQR cost and Riccati equation, local rate feedback, the steady-state Kalman observer, and the control-authority-margin re-tuning of Section V.F — are reproduced in full in the Supplemental Material (Section S4), so that every design is reproducible from the paper and its supplement alone rather than only from the repository.

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[Funding sources and grant numbers, if any – please confirm; this must match what is entered in ScholarOne.]

Data and Code Availability

The plant, baselines, training code, evaluation harness and the SHARPy cross-validation scripts are available at the project repository, including `scripts/animate_flutter.py`, which regenerates the animations referenced in Section III.D for the four classical cases (open loop, centralised LQR, local rate feedback, and centralised LQR with the outboard surface jammed) at the flight condition used throughout the paper; `scripts/robust_baseline_check.py` and `aegis/analysis/decentralized_robustness.py`, which reproduce the control-authority margin analysis of Section V.F; and `scripts/credit_sensitivity.py`, which reproduces the identification-error analysis of Section V.J.

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References

[1] Livne, E., "Aircraft Active Flutter Suppression: State of the Art and Technology Maturation Needs," *Journal of Aircraft*, Vol. 55, No. 1, 2018, pp. 410–452. https://doi.org/10.2514/1.C034442.

[2] Mukhopadhyay, V., "Historical Perspective on Analysis and Control of Aeroelastic Responses," *Journal of Guidance, Control, and Dynamics*, Vol. 26, No. 5, 2003, pp. 673–684. https://doi.org/10.2514/2.5108.

[3] Karpel, M., "Design for Active Flutter Suppression and Gust Alleviation Using State-Space Aeroelastic Modeling," *Journal of Aircraft*, Vol. 19, No. 3, 1982, pp. 221–227. https://doi.org/10.2514/3.57379.

[4] Waszak, M. R., "Robust Multivariable Flutter Suppression for Benchmark Active Control Technology Wind-Tunnel Model," *Journal of Guidance, Control, and Dynamics*, Vol. 24, No. 1, 2001, pp. 147–153. https://doi.org/10.2514/2.4694.

[5] Block, J. J., and Strganac, T. W., "Applied Active Control for a Nonlinear Aeroelastic Structure," *Journal of Guidance, Control, and Dynamics*, Vol. 21, No. 6, 1998, pp. 838–845. https://doi.org/10.2514/2.4346.

[6] Chen, Z., Shi, Z., Chen, S., Tong, S., and Dong, Y., "Active flutter suppression for a flexible wing model with trailing-edge circulation control via reinforcement learning," *AIP Advances*, Vol. 13, No. 1, 2023, p. 015317. https://doi.org/10.1063/5.0130370.

[7] Li, J., Dai, Y., Hu, Y., Zhang, Y., and Yang, C., "Deep reinforcement learning control for stall flutter via active camber morphing," *Physics of Fluids*, Vol. 37, No. 10, 2025, p. 105166. https://doi.org/10.1063/5.0295770.

[8] Mu, X., Huang, R., Zou, Q., and Hu, H., "Machine learning-based active flutter suppression for a flexible flying-wing aircraft," *Journal of Sound and Vibration*, Vol. 529, 2022, p. 116916. https://doi.org/10.1016/j.jsv.2022.116916.

[9] Sunehag, P., Lever, G., Gruslys, A., Czarnecki, W. M., Zambaldi, V., Jaderberg, M., Lanctot, M., Sonnerat, N., Leibo, J. Z., Tuyls, K., and Graepel, T., "Value-Decomposition Networks for Cooperative Multi-Agent Learning," *International Conference on Autonomous Agents and MultiAgent Systems*, 2018. https://doi.org/10.48550/arXiv.1706.05296.

[10] Rashid, T., Samvelyan, M., de Witt, C. S., Farquhar, G., Foerster, J., and Whiteson, S., "QMIX: Monotonic Value Function Factorisation for Deep Multi-Agent Reinforcement Learning," *International Conference on Machine Learning*, 2018. https://doi.org/10.48550/arXiv.1803.11485.

[11] Foerster, J. N., Farquhar, G., Afouras, T., Nardelli, N., and Whiteson, S., "Counterfactual Multi-Agent Policy Gradients," *AAAI Conference on Artificial Intelligence*, Vol. 32, 2018. https://doi.org/10.1609/aaai.v32i1.11794.

[12] Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., and Meger, D., "Deep Reinforcement Learning that Matters," *AAAI Conference on Artificial Intelligence*, 2018. https://doi.org/10.1609/aaai.v32i1.11694.

[13] Engstrom, L., Ilyas, A., Santurkar, S., Tsipras, D., Janoos, F., Rudolph, L., and Madry, A., "Implementation Matters in Deep RL: A Case Study on PPO and TRPO," *International Conference on Learning Representations*, 2020. https://doi.org/10.48550/arXiv.2005.12729.

[14] Andrychowicz, M., Raichuk, A., Stańczyk, P., Orsini, M., Girgin, S., Marinier, R., Hussenot, L., Geist, M., Pietquin, O., Michalski, M., Gelly, S., and Bachem, O., "What Matters for On-Policy Deep Actor-Critic Methods? A Large-Scale Study," *International Conference on Learning Representations*, 2021. https://doi.org/10.48550/arXiv.2006.05990.

[15] Theodorsen, T., "General Theory of Aerodynamic Instability and the Mechanism of Flutter," Tech. Rep. NACA TR-496, National Advisory Committee for Aeronautics, 1935. URL https://ntrs.nasa.gov/citations/19800006788.

[16] Jones, R. T., "The Unsteady Lift of a Wing of Finite Aspect Ratio," Tech. Rep. NACA TR-681, National Advisory Committee for Aeronautics, 1940. URL https://ntrs.nasa.gov/citations/19930091758.

[17] Peters, D. A., Karunamoorthy, S., and Cao, W.-M., "Finite State Induced Flow Models. Part I: Two-Dimensional Thin Airfoil," *Journal of Aircraft*, Vol. 32, No. 2, 1995, pp. 313–322. https://doi.org/10.2514/3.46718.

[18] Fung, Y. C., *An Introduction to the Theory of Aeroelasticity*, John Wiley and Sons, 1955.

[19] Bisplinghoff, R. L., Ashley, H., and Halfman, R. L., *Aeroelasticity*, Addison-Wesley, 1955.

[20] Palacios, R., and Cesnik, C. E. S., *Dynamics of Flexible Aircraft: Coupled Flight Mechanics, Aeroelasticity, and Control*, Cambridge University Press, 2023. https://doi.org/10.1017/9781108354868.

[21] Downs, P. S., and Prazenica, R. J., "Adaptive Control of a Flexible Wing for Flutter Suppression and Disturbance Rejection," *AIAA SCITECH 2023 Forum*, 2023. https://doi.org/10.2514/6.2023-0130.

[22] Mozaffari-Jovin, S., Firouz-Abadi, R. D., and Roshanian, J., "L1 Adaptive Aeroelastic Control of an Unsteady Flapped Airfoil in the Presence of Unmatched Nonlinear Uncertainties," *Journal of Sound and Vibration*, Vol. 578, 2024, p. 118334. https://doi.org/10.1016/j.jsv.2024.118334.

[23] Li, J., Dai, Y., Huang, G., Xia, Y., and Yang, C., "Shapley-Additive-Explanation-Based Stall Flutter Control: from Neural Interpretation to Explainable Policy Design," *AIAA Journal*, 2026, pp. 1–18. https://doi.org/10.2514/1.J066411, published online 16 July 2026.

[24] Shalymov, D., Granichin, O., Ivanskiy, Y., and Volkovich, Z., "Multiagent Control of Airplane Wing Stability with "Feathers" under the Flexural Torsional Flutter," *Mathematics*, Vol. 10, No. 2, 2022, p. 236. https://doi.org/10.3390/math10020236, preprint: arXiv:2012.04582 (2020).

[25] Šiljak, D. D., *Decentralized Control of Complex Systems*, Academic Press, 1991.

[26] Bakule, L., "Decentralized Control: An Overview," *Annual Reviews in Control*, Vol. 32, No. 1, 2008, pp. 87–98. https://doi.org/10.1016/j.arcontrol.2008.03.004.

[27] Rotkowitz, M., and Lall, S., "A Characterization of Convex Problems in Decentralized Control," *IEEE Transactions on Automatic Control*, Vol. 51, No. 2, 2006, pp. 274–286. https://doi.org/10.1109/TAC.2005.860365.

[28] Wolpert, D. H., and Tumer, K., "Optimal Payoff Functions for Members of Collectives," *Advances in Complex Systems*, Vol. 4, No. 2/3, 2001, pp. 265–279. https://doi.org/10.1142/S0219525901000188.

[29] Lowe, R., Wu, Y., Tamar, A., Harb, J., Abbeel, P., and Mordatch, I., "Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments," *Advances in Neural Information Processing Systems*, 2017. https://doi.org/10.48550/arXiv.1706.02275.

[30] Yu, C., Velu, A., Vinitsky, E., Gao, J., Wang, Y., Bayen, A., and Wu, Y., "The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games," *Advances in Neural Information Processing Systems*, 2022. https://doi.org/10.48550/arXiv.2103.01955.

[31] Albrecht, S. V., Christianos, F., and Schäfer, L., *Multi-Agent Reinforcement Learning: Foundations and Modern Approaches*, MIT Press, 2024. Also freely available at https://www.marl-book.com.

[32] Geng, M., Pateria, S., Subagdja, B., and Tan, A.-H., "Scaling Up Multi-Agent Reinforcement Learning for Large Agent Teams and Long-Horizon Tasks: A Survey," *ACM Computing Surveys*, Vol. 58, No. 14, 2026, pp. 1–37. https://doi.org/10.1145/3817113.

[33] Zhao, X., and Xie, Y., "Multi-level Advantage Credit Assignment for Cooperative Multi-Agent Reinforcement Learning," *arXiv preprint arXiv:2508.06836*, 2025. https://doi.org/10.48550/arXiv.2508.06836.

[34] Bernstein, D. S., Givan, R., Immerman, N., and Zilberstein, S., "The Complexity of Decentralized Control of Markov Decision Processes," *Mathematics of Operations Research*, Vol. 27, No. 4, 2002, pp. 819–840. https://doi.org/10.1287/moor.27.4.819.297.

[35] Oliehoek, F. A., and Amato, C., *A Concise Introduction to Decentralized POMDPs*, Springer, 2016. https://doi.org/10.1007/978-3-319-28929-8.

[36] Sukhbaatar, S., Szlam, A., and Fergus, R., "Learning Multiagent Communication with Backpropagation," *Advances in Neural Information Processing Systems*, 2016. https://doi.org/10.48550/arXiv.1605.07736.

[37] Das, A., Gervet, T., Romoff, J., Batra, D., Parikh, D., Rabbat, M., and Pineau, J., "TarMAC: Targeted Multi-Agent Communication," *International Conference on Machine Learning*, 2019. https://doi.org/10.48550/arXiv.1810.11187.

[38] Terry, J. K., Black, B., Grammel, N., Jayakumar, M., Hari, A., Sullivan, R., Santos, L. S., Dieffendahl, C., Horsch, C., Perez-Vicente, R., et al., "PettingZoo: Gym for Multi-Agent Reinforcement Learning," *Advances in Neural Information Processing Systems*, Vol. 34, 2021, pp. 15032–15043. https://doi.org/10.48550/arXiv.2009.14471.

[39] Agarwal, R., Schwarzer, M., Castro, P. S., Courville, A., and Bellemare, M. G., "Deep Reinforcement Learning at the Edge of the Statistical Precipice," *Advances in Neural Information Processing Systems*, 2021. https://doi.org/10.48550/arXiv.2108.13264.

[40] Johannink, T., Bahl, S., Nair, A., Luo, J., Kumar, A., Loskyll, M., Ojea, J. A., Solowjow, E., and Levine, S., "Residual Reinforcement Learning for Robot Control," *IEEE International Conference on Robotics and Automation*, 2019. https://doi.org/10.48550/arXiv.1812.03201.

[41] Silver, T., Allen, K., Tenenbaum, J., and Kaelbling, L., "Residual Policy Learning," *arXiv preprint arXiv:1812.06298*, 2018. https://doi.org/10.48550/arXiv.1812.06298.

[42] Sutton, R. S., and Barto, A. G., *Reinforcement Learning: An Introduction*, 2^{nd} ed., MIT Press, 2018.

[43] Goland, M., "The Flutter of a Uniform Cantilever Wing," *Journal of Applied Mechanics*, Vol. 12, No. 4, 1945, pp. A197–A208. https://doi.org/10.1115/1.4009489.

[44] del Carre, A., Muñoz-Simón, A., Goizueta, N., and Palacios, R., "SHARPy: A dynamic aeroelastic simulation toolbox for very flexible aircraft and wind turbines," *Journal of Open Source Software*, Vol. 4, No. 44, 2019, p. 1885. https://doi.org/10.21105/joss.01885.

[45] Murua, J., Palacios, R., and Graham, J. M. R., "Applications of the Unsteady Vortex-Lattice Method in Aircraft Aeroelasticity and Flight Dynamics," *Progress in Aerospace Sciences*, Vol. 55, 2012, pp. 46–72. https://doi.org/10.1016/j.paerosci.2012.06.001.

[46] Ng, A. Y., Harada, D., and Russell, S., "Policy Invariance Under Reward Transformations: Theory and Application to Reward Shaping," *International Conference on Machine Learning*, 1999.

[47] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O., "Proximal Policy Optimization Algorithms," *arXiv preprint arXiv:1707.06347*, 2017. https://doi.org/10.48550/arXiv.1707.06347.

[48] Schulman, J., Moritz, P., Levine, S., Jordan, M. I., and Abbeel, P., "High-Dimensional Continuous Control Using Generalized Advantage Estimation," *International Conference on Learning Representations*, 2016. https://doi.org/10.48550/arXiv.1506.02438.

[49] Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D., and Wierstra, D., "Continuous Control with Deep Reinforcement Learning," *International Conference on Learning Representations*, 2016. https://doi.org/10.48550/arXiv.1509.02971.

[50] Haarnoja, T., Zhou, A., Abbeel, P., and Levine, S., "Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor," *International Conference on Machine Learning*, 2018. https://doi.org/10.48550/arXiv.1801.01290.

[51] Hausknecht, M., and Stone, P., "Deep Recurrent Q-Learning for Partially Observable MDPs," *AAAI Fall Symposium*, 2015. https://doi.org/10.48550/arXiv.1507.06527.

[52] Anderson, B. D. O., and Moore, J. B., *Optimal Control: Linear Quadratic Methods*, Dover Publications, 2007.

[53] Doyle, J. C., "Guaranteed Margins for LQG Regulators," *IEEE Transactions on Automatic Control*, Vol. 23, No. 4, 1978, pp. 756–757. https://doi.org/10.1109/TAC.1978.1101812.

[54] Doyle, J. C., and Stein, G., "Multivariable Feedback Design: Concepts for a Classical/Modern Synthesis," *IEEE Transactions on Automatic Control*, Vol. 26, No. 1, 1981, pp. 4–16. https://doi.org/10.1109/TAC.1981.1102555.